Humanizing Machine-Generated Content: Evading AI-Text Detection through Adversarial Attack

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Abstract

With the development of large language models (LLMs), detecting whether text is generated by a machine becomes increasingly challenging in the face of malicious use cases like the spread of false information, protection of intellectual property, and prevention of academic plagiarism. While well-trained text detectors have demonstrated promising performance on unseen test data, recent research suggests that these detectors have vulnerabilities when dealing with adversarial attacks such as paraphrasing. In this paper, we propose a framework for a broader class of adversarial attacks, designed to perform minor perturbations in machine-generated content to evade detection. We consider two attack settings: white-box and black-box, and employ adversarial learning in dynamic scenarios to assess the potential enhancement of the current detection model's robustness against such attacks. The empirical results reveal that the current detection models can be compromised in as little as 10 seconds, leading to the misclassification of machine-generated text as human-written content. Furthermore, we explore the prospect of improving the model's robustness over iterative adversarial learning. Although some improvements in model robustness are observed, practical applications still face significant challenges. These findings shed light on the future development of AI-text detectors, emphasizing the need for more accurate and robust detection methods.

1 Introduction

Large language models (LLMs) (OpenAI, 2023; Anil et al., 2023; Touvron et al., 2023) have rapidly emerged as a dominant force within the field of natural language processing (NLP). These models acquire extensive internal knowledge through pre-training on large-scale self-supervised data, endowing them the capacity to tackle various tasks, from answering factual questions to generating fluent text, and even performing complex reasoning, which has significantly impacted diverse NLP application domains. However, these advancements have also raised ethical concerns on their inherent risks (McKenna et al., 2023; Bian et al., 2023; Ferrara, 2023), including the spread of misinformation, the hallucinations in generated content, and even potential discrimination against specific groups. The growing recognition of these issues has led to the development of AI-text detection research. Nevertheless, AI-text detector may inherit vulnerabilities from neural network models (Szegedy et al., 2014), spurring related research (Sadasivan et al., 2023; Krishna et al., 2023) aimed at conducting paraphrasing attacks on AI detectors to mislead their predictions. We believe that the exploration of potential adversarial attacks on text detectors is of paramount importance, as it allows for the identification of vulnerabilities in AI detectors before their deployment in real-world applications, such as student essay plagiarism detection, while also facilitating the development of appropriate countermeasures.

Current detection methods are typically categorized into three groups: those relying on statistical measures (Mitchell et al., 2023) like entropy, perplexity, and log-likelihood; those training neural classifiers (Guo et al., 2023) from supervised data with human/AI-generated labels; and those utilizing watermarking (Kirchenbauer et al., 2023) to inject imperceptible pattern to the AI-generated text. Unfortunately, limited research has explored the adversarial perturbations targeting AI-text detectors. Notably, Sadasivan et al. (2023); Krishna et al. (2023) explored the use of paraphraser to rewrite machine-generated content for adversarial attacks. Simultaneously, Shi et al. (2023) utilized LLMs to generate adversarial word candidates, and created adversarial results using a search-based method. While these prior studies have revealed the vulnera-

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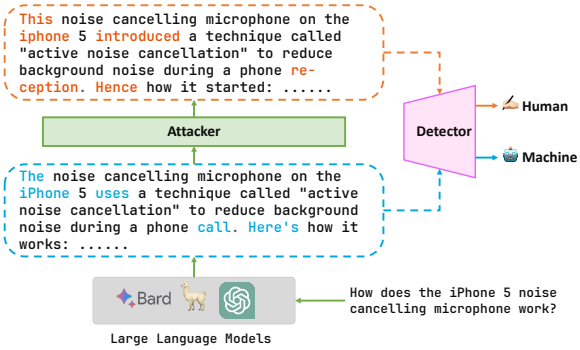

Figure 1: Overview of the Adversarial Detection Attack on AI-Text (ADAT) task.

bilities of AI detectors to adversarial perturbations, the influence of adversarial attacks on the detector in real-world dynamic scenarios remains largely unexplored.

In this paper, we propose a broader task: Adversarial Detection Attack on AI-Text (ADAT). The objective of ADAT is to perturb AI-generated text in a semantically preserving manner, thereby influencing the detector's predictions and enabling machine-generated text to evade detection. Figure 1 outlines the general process of ADAT, along with an example of attacks[1]. We consider both black-box attack setting, where the attacker can access only the target detector predictions without any internal information of the detection model, and white-box attack setting. To bridge the gap between these settings, we employ an intermediate model to unify the attack methods. Additionally, to address the challenge of constructing a more robust detection model, we introduce an adversarial learning approach in a dynamic scenario for the ADAT task, whereas the detector iteratively updates its parameters using adversarial samples. Our experiments validate the enhanced robustness achieved through this approach and highlight potential challenges. Building on the above considerations, we introduce a novel and comprehensive framework called the Humanizing Machine-Generated Content (HMGC) for ADAT tasks, which is designed to facilitate the interaction process between the attacker and the detector.

Our main contributions could be summarized as follows:

- To the best of our knowledge, ADAT is the first rigorously defined task in the field of adversarial attacks on AI-text detection. It encompasses both white-box and black-box attack settings, serving as a foundational reference for future research in this domain.
- We introduce the HMGC framework, which offers a general attack paradigm for ADAT tasks. Extensive experiments reveal the efficacy of the HMGC framework. In particular, we provide empirical evidence highlighting the significance of perplexity in AI-text detection.
- Our experimental results demonstrate that the proposed adversarial learning approach in dynamic scenarios effectively enhances the robustness of detection models, suggesting the potential to train a universal AI-text detector through dynamic adversarial learning. The data and related resources are available online[2].

2 Related Works

Large Language Model. The powerful capabilities of large language models, exemplified by GPT (OpenAI, 2023), PaLM (Anil et al., 2023), and LLaMA (Touvron et al., 2023) have revolutionized the application landscape within the field of natural language processing. These models can generate coherent and fluent text enriched with external knowledge, effectively tackling complex issues across various domains, from physics (West, 2023) and medicine (DiGiorgio and Ehrenfeld, 2023) to mathematics (Li et al., 2023) and linguistics (Liu et al., 2023b). Nonetheless, current language models still grapple with issues like hallucination (McKenna et al., 2023), the inadvertent spread of misinformation (Bian et al., 2023), and the potential for value discrimination (Ferrara, 2023) in practical applications. Consequently, the regulation of large language models to mitigate the risk of significant social problems has become increasingly vital. We focus on studying how AI-generated text can circumvent existing detection mechanisms, aiming to provide valuable insights and perspectives for the development of robust detection models.

AI-text Detection. Current text detection methods typically fall into three categories: 1) Statistical methods (Gehrmann et al., 2019; Lavergne et al., 2008; Solaiman et al., 2019; Mitchell et al., 2023; Su et al., 2023) employ statistical tools to make zero-shot distinctions between human and machine-generated text using measures such as information

[1]Sample in figure 1 is from real AI-text attacks, available for testing at https://huggingface.co/Hello-SimpleAI/chatgpt-detector-roberta

[2]https://github.com/zhouying20/HMGC

entropy, perplexity, and n-gram frequency. Recent work in this category includes DetectGPT (Mitchell et al., 2023), which observed that text generated by language models often resides in the negative curvature region of the log probability function, and it proposed the defining curvature-based criteria to distinguish AI-text. 2) Classifier-based methods involve training text detection models based on supervised data (Uchendu et al., 2020; Deng et al., 2023; Mireshghallah et al., 2023), whereas recent research (Guo et al., 2023; Liu et al., 2023c) often utilizes RoBERTa to train a binary classifier for text detection. However, this method requires a substantial amount of training data and faces challenges related to weak generalization (Bakhtin et al., 2019) and limited robustness against attacks (He et al., 2023; Qi et al., 2021). 3) Watermarking methods are emerging with the rise of decoder-only large language models, which imprint specific patterns on generated text. For instance, Kirchenbauer et al. (2023) propose randomly partitioning the vocabulary into a green list and a red list during text generation, with the division based on the hash of previously generated tokens. Meanwhile, Liu et al. (2023a) introduces a watermarking method akin to the RSA asymmetric cryptography. Challenges for this category include that the generated text may lack smoothness, and the watermark pattern is susceptible to leakage (Liu et al., 2023a).

Adversarial Attack. A few recent works have made attempts to attack text detection models. For example, Sadasivan et al. (2023); Krishna et al. (2023) proposed to use paraphrasers to rewrite the generated text of LLMs, successfully evading detection by the 3 categories of models mentioned above. Notably, our work shares common views on the usage of adversarial learning with Hu et al. (2023), but differs in that Hu et al. (2023) introduced a paraphraser to enhance the robustness of detector, whereas our work explores whether the detector can continue to learn from multiple rounds of attacks in dynamic scenarios to resist adversarial attacks. Shi et al. (2023) demonstrated the utility of word substitution attacks against AI-text detectors. When compared to our approach, it's worth noting that their method relies on LLMs to generate candidate words, which might encounter efficiency challenges when computing resources are limited. Building upon these observations, our paper presents a comprehensive detector adversarial attack framework called HMGC. The key distinction between our work and these studies lies in our introduction of the ADAT task, providing a formal paradigm for future research. Moreover, we also emphasize our work as universal adversarial perturbations, which can be applied to any input for any target detector model.

3 Preliminary

In this section, we introduce the key definitions of the Adversarial Detection Attack on AI-Text (ADAT) task.

3.1 Problem Statement

In the detection of machine-generated content, when presented with a set of user requirements $\mathcal{U}$ and their corresponding response articles $\mathcal{T} = \mathcal{T}_{human} \cup \mathcal{T}_{machine}$, the objective of the detection model $\mathcal{D}$ is to assign a score $\mathcal{D}(u, t)$ to each article to help users discern whether the article is generated by machines, specifically for large language models (LLMs) like ChatGPT or Bard. For instance, given a threshold of 0.5, the detector generates detection probabilities for all articles, resulting in an array $\mathcal{P} = [D_1, D_2, ..., D_n]$, in which articles with a detection probability exceeding 0.5 are identified as machine-generated content, defined as $\overline{\mathcal{T}}_{machine} = \{t \mid D(u_t, t) > 0.5\}$.

Adversarial Attack on Detection. We propose the task of Adversarial Detection Attack on AI-Text (ADAT). The objective is to introduce subtle modifications to machine-generated articles, aiming to fool the detector into classifying them as human-authored. Formally defined, given user requirements u_t and a machine-generated article t, the attack's objective is to construct an effective adversarial sample t_{adv} based on t, ensuring its detection probability falls below the classification threshold. Concretely, we define a successful attack as: $D(u_t, t_{adv}) < threshold$ when $D(u_t, t) \geq threshold$, with the condition that $Dis(t, t_{adv}) \leq \epsilon$. Here, $D(u_t, t)$ and $D(u_t, t_{adv})$ are the detection probabilities of the text before and after the attack, respectively. Dis is the similarity distance evaluation function, and ϵ is a small value ensuring minimal deviation of the text distribution in the adversarial sample from the original text.

White-box and Black-box Attack. In terms of attack methodologies against the detector, we establish two realistic settings: In the **white-box attack** scenario, the attacker accesses comprehensive information about the victim detector, includ-